Hot Topic

Inequalities

You can use these symbols to describe **inequalities**. The fat end of the symbol always points towards the bigger number.

$>$ means 'is greater than'

$\geqslant$ means 'is greater than or equal to'

$<$ means 'is less than'

$\leqslant$ means 'is less than or equal to'

Worked example

Aron rounds a number, n, to 2 decimal places. The result is 8.64.
Using inequalities, write down the error interval for n. **(2 marks)**

$8.635 \leqslant n < 8.645$

Error intervals

You can use inequalities to show the error interval when a number has been **rounded**:

1. The height of the London Eye is 140 m (to the nearest 10 m). You can write $135\text{ m} \leqslant \text{Height} < 145\text{ m}$

The least possible value for the height is 135 m.

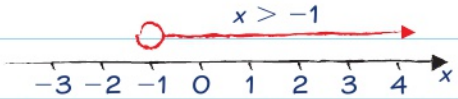
You use 'less than' for the greatest possible value.

2. $x = 4.7$ (1 d.p.) means $4.65 \leqslant x < 4.75$

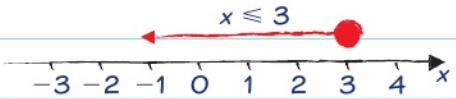
You use 'less than or equal to' for the least possible value.

Inequalities on number lines

You can represent inequalities on a number line.



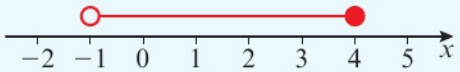
The open circle shows that −1 is **not** included.



The closed circle shows that 3 **is** included.

Worked example

(a) Show the inequality $-1 < x \leqslant 4$ on the number line below.



(1 mark)

(b) x is an integer and $-1 < x \leqslant 4$. Write down all the possible values of x.

0, 1, 2, 3, 4 **(2 marks)**

Examiners' report

Integers are positive and negative whole numbers, including zero. So you **should** include zero in your answer to part (b).
Look at the inequalities carefully.

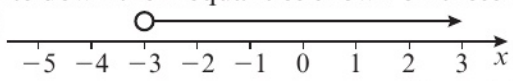
- The bottom inequality is $<$ so **don't** include -1.
- The top inequality is $\leqslant$ so **do** include 4.

Real students have struggled with questions like this in recent exams – **be prepared!**

Now try this

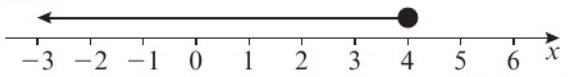
1 Write down the inequalities shown on these number lines.

(a)



(1 mark)

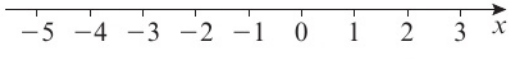
(b)



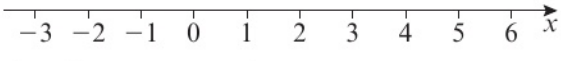
(1 mark)

2 Show these inequalities on the number lines below.

(a) $-4 < x < 1$ **(1 mark)**



(b) $-2 \leqslant x < 5$ **(1 mark)**



Solving inequalities

You can solve an inequality in exactly the same way as you solve an equation.

$2x - 3 \leqslant 15$ $(+ 3)$
$2x \leqslant 18$ $(\div 2)$
$x \leqslant 9$

Watch out!

If you multiply or divide by a **negative number** when solving an inequality, you have to **reverse the direction** of the inequality sign. It's usually easier to eliminate negative terms by **adding**.

Solutions

The solution has the letter on its own on one side of the inequality and a number on the other side.

Solutions to inequalities
$x > 4$ $-2 < x$ $x \leqslant -\frac{3}{4}$

Not solutions to inequalities
$x \geqslant 20 + 3$ $2x < 10$ $x = 4$

Worked example

Solve the inequality

(a) $2x + 1 < 10$ $(- 1)$ **(2 marks)**
$2x < 9$ $(\div 2)$
$x < \frac{9}{2}$ or $x < 4\frac{1}{2}$

(b) $5x < 2x - 6$ $(- 2x)$ **(2 marks)**
$3x < -6$ $(\div 3)$
$x < -2$

(c) $12 - 4x \geqslant 10$ $(+ 4x)$ **(3 marks)**
$12 \geqslant 10 + 4x$ $(- 10)$
$2 \geqslant 4x$ $(\div 4)$
$\frac{2}{4} \geqslant x$ or $\frac{1}{2} \geqslant x$

These are **inequalities** and not equations. You don't need to use an = sign anywhere in your answer. Write down the operation you are using at each step and remember you have to do the same thing to both sides of the inequality. Your answer can involve negative numbers or fractions.

When you're solving an inequality you should avoid multiplying or dividing by a **negative** number. Add $4x$ to both sides of the equation to make the x term positive.

Worked example

Write down all the integers that satisfy the inequality:
$-15 \leqslant 5n < 10$ **(3 marks)**

$-15 \leqslant 5n$ $5n < 10$ $(\div 5)$
$-3 \leqslant n$ $n < 2$
$n = -3, -2, -1, 0, 1$

This is really two inequalities:
$-15 \leqslant 5n$ and $5n < 10$
Solve them separately, then write down all the integers which satisfy **both** inequalities at the same time.

Now try this

1 Solve the following inequalities.
(a) $4x + 3 \geqslant 23$ **(2 marks)**
(b) $18 - 5x > -2$ **(3 marks)**

2 Find all the integers which satisfy each of these inequalities.
(a) $-8 \leqslant 4x < 12$ **(3 marks)**
(b) $-1 < 3x \leqslant 10$ **(3 marks)**

Worked solution video